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Methods Of Coincidence Measurement and Electron-Positron Annihilation

[ALC - B19]

M766

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Experiment Date: 09.07.2025

Abstract

This report outlines the theoretical foundations and experimental setup for a coincidence measurement designed to study electron-positron annihilation using a ^{22}Na source. The experiment focuses on detecting simultaneously emitted γ -rays resulting from β^+ decay, utilizing scintillation detectors coupled with photomultiplier tubes (PMTs) and a series of timing electronics including threshold discriminators, time-to-pulse-height converters (TPC), and single-channel analyzers (SCA). The goal is to understand the principles behind isolating true coincidence events from random background, and to prepare for the calibration of time and energy spectra.

A significant part of the experiment will involve examining positron annihilation in a magnesium sample to explore the angular correlation of the 511 keV γ -rays. This measurement will provide the basis for estimating the Fermi momentum and Fermi energy of conduction electrons in metals. In addition, concepts such as photon detection efficiency, timing resolution, and statistical treatment of random coincidences will be addressed. This report serves as the theoretical and procedural groundwork for the upcoming measurements.

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1 Introduction

1.1 Introduction

Coincidence measurements are a fundamental technique in experimental particle physics, particularly valuable for studying reactions that produce two or more correlated particles. By requiring that detected signals occur within a narrow time window, this method effectively filters out uncorrelated background events and isolates interactions of physical interest. Such techniques are widely implemented in high-energy physics experiments at particle accelerators and also serve as the foundation for medical imaging technologies like Positron Emission Tomography (PET), where simultaneous detection of annihilation photons is crucial.

In the context of this experiment, the coincidence method will be applied to study the annihilation of positrons emitted during the β^+ decay of a ^{22}Na source. The annihilation produces two 511 keV γ -rays emitted approximately back-to-back due to momentum conservation. To detect these coincident photons, the experimental setup includes scintillation detectors based on NaI(Tl) crystals, coupled to photomultiplier tubes (PMTs), and supported by a chain of electronic modules such as threshold discriminators, time-to-pulse-height converters (TPC), and single-channel analyzers (SCA). These components work together to process the detector signals and identify truly simultaneous events.

A further focus of the experiment involves investigating electron-positron annihilation in a magnesium target. The goal is to analyze the angular distribution of coincident γ -quanta and, from this, derive properties of the conduction electrons in the metal, such as the Fermi momentum and Fermi energy. This serves as a practical application of the coincidence method in solid-state physics and offers insight into the momentum distribution of electrons in metals.

This report provides the theoretical background and outlines the experimental procedure in preparation for performing the actual measurements.

2 Theoretical Background

Before performing the experiment, it is essential to understand the fundamental physics behind β^+ decay, positron annihilation, and the angular correlation of the resulting γ -quanta. These principles are crucial for interpreting the measurements and evaluating the performance of the coincidence detection setup.

2.1 Beta-Plus Decay and Positron Annihilation

2.1.1 Weak Interaction and the Origin of Positrons

In β^+ decay, an unstable nucleus reduces its proton-to-neutron ratio by converting a proton into a neutron. This transformation occurs through the weak nuclear force and is mediated by the emission of a virtual W^+ boson, which subsequently decays into a positron (e^+) and an electron neutrino (ν_e). The full reaction is expressed as:

$$p \rightarrow n + e^+ + \nu_e.$$

This process alters the atomic number of the nucleus while conserving charge and lepton number. On the subatomic level, the transition involves the conversion of an up quark into a down quark within the proton, marking the change to a neutron. Since a free proton is lighter than a free neutron, such a decay is only energetically feasible when the proton is bound in a nucleus with sufficiently high binding energy to overcome the mass difference [9, 7].

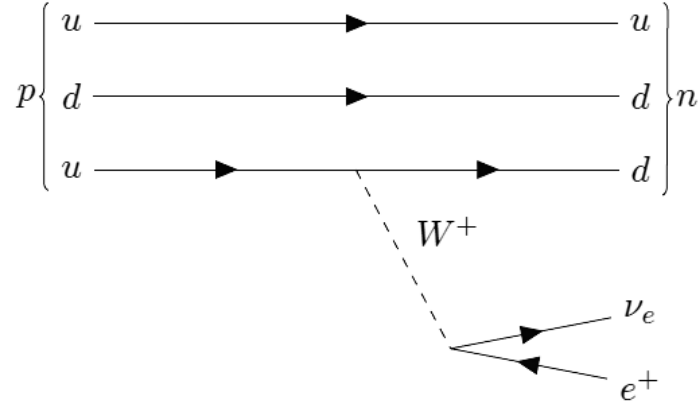


Figure 2.1: Feynman diagram of β^+ decay at the quark level. An up quark in the proton converts to a down quark via the weak interaction, emitting a W^+ boson which decays into a positron and an electron neutrino. [4].

2.1.2 Energy Loss and Thermalization of Positrons

Once emitted, the positron interacts with surrounding matter, primarily through ionization and scattering processes. These interactions lead to a rapid loss of kinetic energy, with the positron thermalizing in about 10^{-10} seconds. The distance over which this occurs is relatively short, as the positron slows down and approaches rest. During this time, it may form a short-lived bound state with an electron known as positronium, particularly in low-density or non-metallic environments. However, in metals, direct annihilation is more common due to the high density of conduction electrons and rapid momentum exchange [5].

2.1.3 Electron-Positron Annihilation and Photon Emission

When a thermalized positron encounters an electron, they annihilate. The most probable channel for annihilation is the emission of two photons, each with an energy of 511 keV, corresponding to the rest mass of the electron or positron. This occurs when the positron and electron form a singlet state (with antiparallel spins), which allows for annihilation into two photons. The process conserves energy and momentum and is represented by:

$$e^+ + e^- \rightarrow \gamma + \gamma.$$

This reaction dominates due to its lower order in quantum electrodynamics compared to other, less probable channels such as three-photon annihilation, which can occur in the triplet spin state but is suppressed by a factor proportional to the fine-structure constant α [8].

2.1.4 Angular Correlation and 180° Emission

In the ideal case where the electron and positron are both at rest (or their net momentum is negligible), the two emitted γ -rays must be emitted back-to-back, i.e., at an angle of 180°. This strict angular correlation arises from the conservation of linear momentum in the center-of-mass frame of the annihilating pair. Since the total momentum before annihilation is zero, the resulting momenta of the photons must be equal in magnitude and opposite in direction. This property is a cornerstone of coincidence detection and is widely exploited in applications such as positron emission tomography (PET) [7].

2.1.5 Deviation from 180° in Metallic Environments

While the 180° emission is expected in vacuum or insulating materials, deviations can occur in metallic environments due to the properties of conduction electrons. In metals, electrons are delocalized and follow a Fermi-Dirac distribution. Their momenta span a range up to the Fermi momentum p_F , as described by the free electron gas model. When a positron annihilates with a conduction electron carrying non-zero momentum, the center-of-mass of the pair is no longer at rest. This results in the two photons being emitted at an angle slightly less than 180°, forming a narrow angular distribution centered around the ideal value [6].

This deviation is not just a subtle correction—it serves as a measurable quantity that reflects the underlying momentum distribution of electrons in the material. In this experiment, a magnesium sample is used to study this effect. By analyzing the angular spread of the annihilation photons, it becomes possible to estimate the Fermi momentum p_F and, consequently, the Fermi energy E_F of the conduction electrons. This represents a powerful intersection of particle physics and solid-state physics, enabling structural insights through annihilation measurements.

2.2 Interaction of γ -Radiation with Matter

2.2.1 Nature and Origin of γ -Radiation

Gamma radiation (γ -radiation) is a form of high-energy, electromagnetic radiation typically emitted during nuclear transitions. Unlike α or β radiation, γ rays do not carry charge or mass, but they possess significant energy, usually in the range of tens of keV to several MeV. These photons are released when a nucleus undergoes de-excitation from a higher energy state to a lower one, often following β decay or nuclear capture processes [9, 7].

Common sources of γ -radiation include:

- Natural radioactive isotopes (e.g., ^{40}K , ^{238}U series),

- Artificial radionuclides used in medical and laboratory applications (e.g., ^{60}Co , ^{137}Cs , ^{22}Na),
- Astrophysical events such as gamma-ray bursts and solar flares.

In this experiment, the γ -radiation originates from the decay of ^{22}Na , a commonly used positron-emitting isotope. The decay scheme involves β^+ emission to an excited state of ^{22}Ne , which then de-excites via the emission of a 1275 keV γ -photon. Additionally, the emitted positron annihilates with an electron, producing two 511 keV γ -quanta. Thus, the decay spectrum of ^{22}Na features two prominent γ -lines: one at 511 keV and another at 1275 keV. The former is a direct result of electron-positron annihilation, while the latter corresponds to the nuclear de-excitation of $^{22}\text{Ne}^*$ [5, 8].

2.2.2 Mechanisms of γ -Ray Interaction with Matter

When γ -rays pass through matter, they interact with atomic electrons or nuclei primarily through three dominant processes, depending on the energy of the incoming photon and the atomic number of the absorbing medium:

1. **Photoelectric Effect:** This process dominates at lower photon energies (typically below 100 keV). A photon is completely absorbed by an atom, transferring all its energy to a tightly bound electron, which is then ejected. The probability of this effect scales with approximately Z^5 and inversely with $E_\gamma^{3.5}$, making it highly sensitive to both atomic number and photon energy [7].
2. **Compton Scattering:** In the intermediate energy range (100 keV to several MeV), Compton scattering becomes the most probable interaction. Here, the γ -photon undergoes inelastic scattering with a weakly bound or free electron, transferring part of its energy to the electron and being deflected in direction. The scattered photon has reduced energy, and the recoil electron carries the rest. The cross-section scales approximately linearly with atomic number ($\sigma \propto Z$) and decreases with increasing energy [5].
3. **Pair Production:** At photon energies exceeding 1.022 MeV (twice the rest mass energy of an electron), the γ -photon can transform into an electron-positron pair when interacting with the electromagnetic field of a nucleus. This process becomes increasingly significant at high energies and is more likely in materials with large atomic numbers ($\sigma \propto Z^2 \ln E_\gamma$). The positron eventually annihilates, producing additional secondary photons [9].

The relative importance of these mechanisms is summarized in Figure 2.2 (not shown here), and the attenuation of γ -radiation is described by the exponential law:

$$I(x) = I_0 \cdot e^{-\mu x},$$

where I_0 is the initial intensity, x is the absorber thickness, and μ is the linear attenuation coefficient. The quantity μ depends on the total contribution from all interaction mechanisms and is material- and energy-dependent [7].

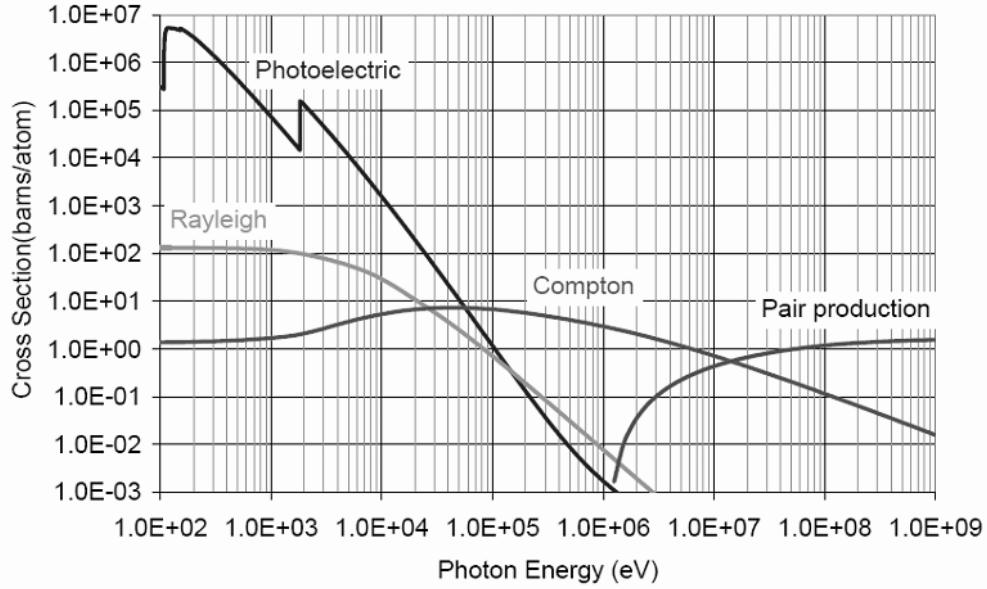


Figure 2.2: Dominant interaction mechanisms of γ -radiation with matter as a function of photon energy. The plot shows the cross-sections for the photoelectric effect, Compton scattering, and pair production across several orders of magnitude in energy. Adapted from [2].

2.2.3 Significance for Detector Response

Understanding these interaction channels is crucial for interpreting the energy spectra recorded by the scintillation detectors. In the energy range relevant to ^{22}Na decay, both photoelectric and Compton effects contribute significantly to the spectrum. The full-energy peaks at 511 keV and 1275 keV are associated with total energy deposition via the photoelectric effect or successive Compton interactions. The continuous background between the peaks results from partial energy losses via Compton scattering, where the scattered photon escapes the detector volume.

Accurate modeling of these interactions allows for proper calibration of the energy scale, estimation of detector efficiency, and analysis of angular correlations between coincident photons.

2.3 Scintillation Detectors and Photomultiplier Tubes

2.3.1 Detection of γ -Radiation

As previously discussed, γ -radiation interacts with matter via the photoelectric effect, Compton scattering, and pair production. These interactions occur within the detecting medium and lead to the production of energetic electrons. The energy deposited by these electrons is the key quantity that can be measured in a radiation detector.

The exact contribution of each interaction type depends strongly on the photon energy and the atomic number Z of the detector material. For instance:

- Below 100 keV, the **photoelectric effect** dominates.
- Between 100 keV and a few MeV, **Compton scattering** is most significant.
- Above 1.022 MeV, **pair production** becomes increasingly relevant.

This energy-dependent behavior shapes the form of a γ -ray energy spectrum, as shown in Figure 2.4.

In a typical spectrum from a source like ^{22}Na , one observes:

- A **Compton continuum** resulting from scattered electrons depositing partial energy.
- **Full-energy peaks** at specific energies (511 keV and 1275 keV), where the entire photon energy is absorbed, usually via photoelectric effect or complete energy transfer through multiple interactions.

2.3.2 Scintillation Detectors: Principle and Operation

Scintillation detectors are widely used in experimental nuclear and particle physics for detecting ionizing radiation, including γ -rays. They operate based on the conversion of deposited energy into visible light, which is subsequently amplified and digitized.

A typical scintillation detector consists of:

- A **scintillator crystal**, such as NaI(Tl), which emits visible photons when excited by incoming radiation.
- A **photomultiplier tube (PMT)** or photodiode, which converts this light into an electrical signal.

When a γ -photon enters the crystal, it undergoes one of the aforementioned interaction mechanisms, producing a high-energy electron. This electron then loses its energy by exciting the atoms of the scintillator, which respond by emitting low-energy (visible) photons—a process known as *scintillation*. The intensity of the emitted light is proportional to the energy deposited by the radiation [5, 7].

2.3.3 Photomultiplier Tubes (PMTs): Signal Amplification

The scintillation light is typically weak and requires amplification to be measurable. A photomultiplier tube (PMT) accomplishes this through a chain of secondary emission processes:

1. The visible photons from the scintillator strike the **photocathode**, which emits electrons via the photoelectric effect.
2. These electrons are accelerated toward a series of **dynodes** held at increasingly positive potentials.
3. At each dynode, multiple secondary electrons are emitted, resulting in a cascade that amplifies the original signal by a factor of 10^5 to 10^6 .
4. The final electron pulse is collected at the **anode** and converted into a voltage signal.

This voltage signal is then processed by timing and discrimination electronics for further analysis, such as energy calibration or coincidence measurement.

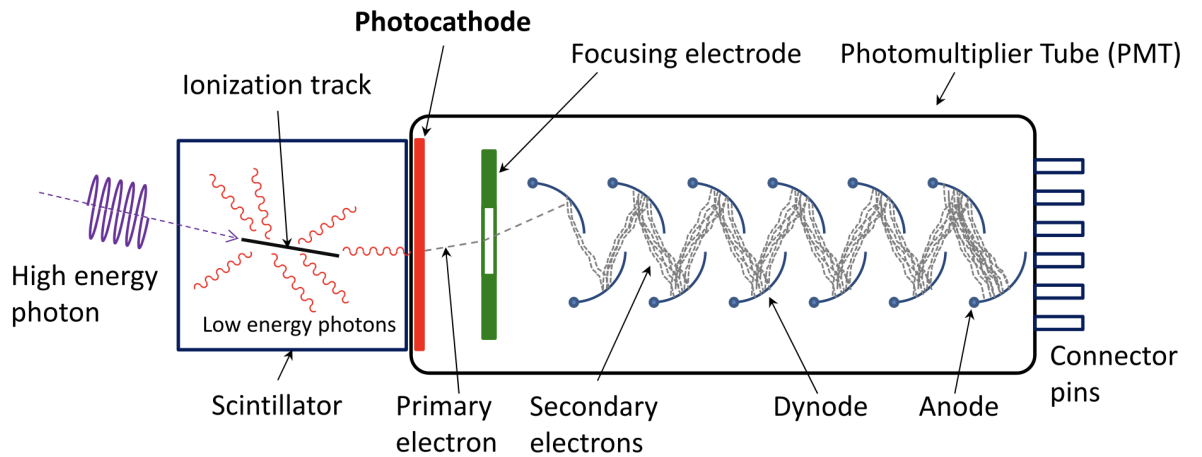


Figure 2.3: Schematic representation of a scintillation detector with photomultiplier tube (PMT). A high-energy photon interacts in the scintillator material, producing an ionization track that leads to the emission of low-energy scintillation photons. These are detected by the photocathode and converted into primary electrons. The electrons are then multiplied through a cascade across a series of dynodes, resulting in a strong electrical pulse at the anode. Adapted from [3].

2.3.4 Energy Spectra and Detector Response

The electrical output of the detector is proportional to the energy deposited by the radiation. When connected to an analog-to-digital converter (ADC), the output can be used to generate a histogram representing the energy distribution of detected events—this is the energy spectrum.

For example, a ^{22}Na source produces:

- A full-energy peak at 511 keV from positron annihilation,
- A second peak at 1275 keV from nuclear de-excitation,
- And a broad Compton background between them, reflecting partial energy losses due to Compton scattering.

The height and width of the peaks, and the relative intensities of features in the spectrum, depend on the scintillator's resolution, geometry, and the dominant interaction mechanisms at each energy level.

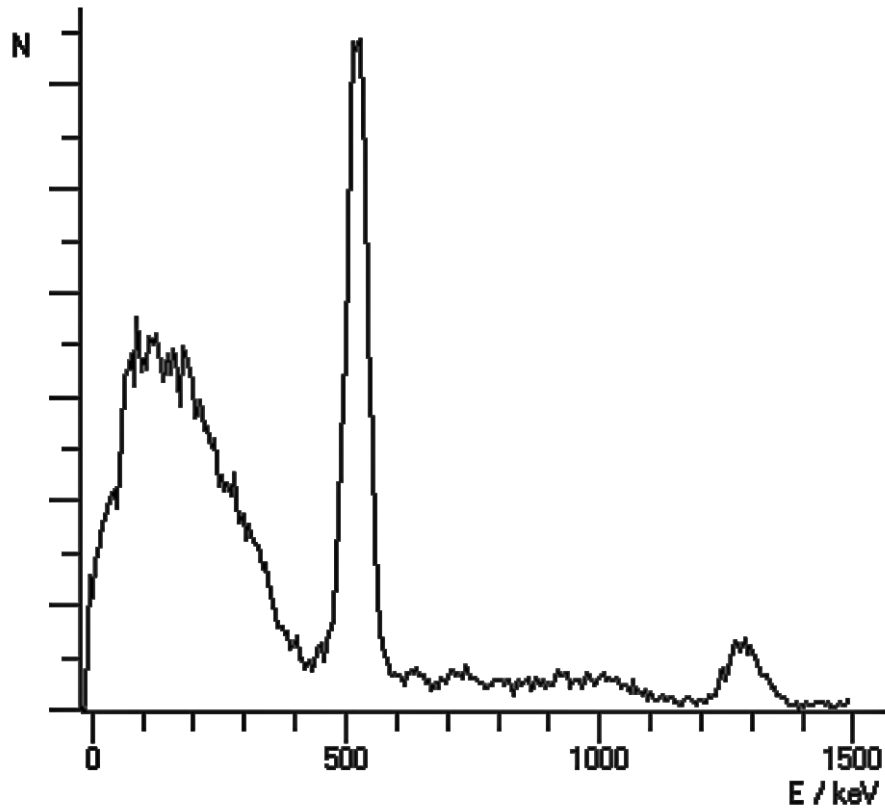


Figure 2.4: Typical γ -ray energy spectrum of a ^{22}Na source, showing the Compton continuum, the annihilation peak at 511 keV, and the de-excitation peak at 1275 keV from the transition of $^{22}\text{Ne}^*$ to its ground state. Adapted from [10].

2.4 Fermi Energy and the Fermi Gas Model

2.4.1 Introduction to the Fermi Gas Concept

In solid-state physics, particularly in the study of metals, electrons can be modeled as a quantum mechanical gas of non-interacting fermions. This so-called *free electron Fermi gas* assumes that conduction electrons move freely within a potential well formed by the periodic lattice of positive ions. Despite the apparent simplicity, this model captures essential features of electron behavior in metals and serves as the foundation for understanding phenomena like electrical conductivity, heat capacity, and electron momentum distributions [6].

Because electrons are fermions, they obey the Pauli exclusion principle, which prohibits multiple electrons from occupying the same quantum state. As a result, at absolute zero temperature ($T = 0\text{ K}$), the available energy states are filled up to a maximum energy called the *Fermi energy*, E_F . All states below E_F are occupied, and all states above are empty.

2.4.2 Fermi-Dirac Distribution and Temperature Dependence

The probability that an energy state E is occupied by an electron at temperature T is given by the Fermi-Dirac distribution function:

$$f(E) = \frac{1}{\exp\left(\frac{E-E_F}{k_B T}\right) + 1},$$

where k_B is the Boltzmann constant, and E_F is the Fermi energy [11].

At $T = 0\text{ K}$, this distribution becomes a step function:

$$f(E) = \begin{cases} 1, & E < E_F \\ 0, & E > E_F \end{cases}$$

This implies that all energy levels up to E_F are fully occupied, while higher levels are empty. As the temperature increases, the step becomes smoother, and some electrons gain enough thermal energy to occupy states slightly above the Fermi energy.

The shape of this distribution is fundamental in determining the behavior of conduction electrons in metals, especially in phenomena involving electron transport and scattering.

2.4.3 Definition of the Fermi Energy

The Fermi energy E_F is defined as the energy of the highest occupied electron state at absolute zero. It depends on the electron density and is given by the expression:

$$E_F = \frac{\hbar^2}{2m_e} \left(3\pi^2 n \right)^{2/3},$$

as derived for a three-dimensional free electron gas at zero temperature [1].

where:

- $\hbar$ is the reduced Planck constant,
- m_e is the electron mass,
- n is the number density of conduction electrons.

For most metals, E_F lies in the range of 2 to 10 eV. This energy sets the scale for many electronic properties of the material. The corresponding *Fermi momentum* is:

$$p_F = \sqrt{2m_e E_F},$$

which represents the maximum momentum of electrons at $T = 0$ K.

2.4.4 Relevance in Positron Annihilation

In the context of this experiment, the Fermi energy plays a crucial role in explaining deviations from ideal back-to-back photon emission in positron-electron annihilation. In metals, the electrons are not stationary but possess momentum up to p_F . When a positron annihilates with a conduction electron, the non-zero momentum of the electron results in an angular deviation of the 511 keV γ -rays from the ideal 180° configuration.

By experimentally measuring this angular spread, one can estimate the Fermi momentum and, consequently, the Fermi energy of the electrons in the metal. This approach provides an indirect method to probe the electronic structure of materials like magnesium, which is used in this experiment.

3 Experimental Setup

3.1 Principle of Coincidence Measurements

Coincidence detection is a powerful technique employed to selectively measure correlated events that occur nearly simultaneously in two independent detectors. In the context of positron annihilation, two photons are ideally emitted in opposite directions. If both detectors are aligned appropriately and an event is registered in each within a short time interval (typically in the nanosecond range), it is highly likely that the detections originated from the same annihilation event. This allows for the rejection of uncorrelated background and improves measurement sensitivity.

Such measurements are fundamental to various applications, including nuclear spectroscopy and medical imaging (e.g., PET), where the detection of simultaneous γ -photons reveals spatial or temporal correlations between decay products. In this experiment, the coincidence technique is utilized to study angular correlations of 511 keV annihilation photons, and to isolate them from other radiation components present in the ^{22}Na decay.

3.2 Setup Description and Geometry

The experimental configuration consists of two NaI(Tl) scintillation detectors placed on opposite sides of a ^{22}Na source. Each detector is coupled to a photomultiplier tube (PMT) to convert light flashes from the scintillator into measurable electrical pulses. The detectors can be rotated to specific angles relative to the source, enabling angular correlation studies. During the initial phase, the detectors are fixed face-to-face at 180° , with a separation of 5 cm from the source.

Signals from the detectors are processed by a series of electronic components that distinguish coincident from random events. A schematic of the entire measurement setup is shown in Figure 3.1. The circuit is designed to analyze both the energy and timing of incoming pulses to identify valid photon pairs.

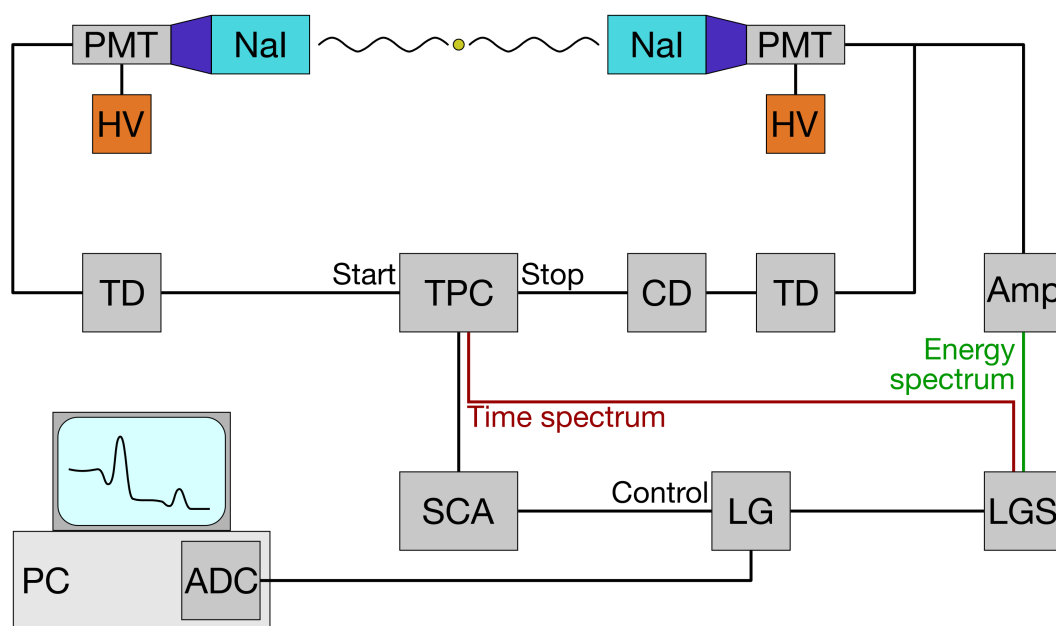


Figure 3.1: Schematic of the measurement setup including all necessary components of the coincidence circuitry. Adapted from [12].

3.3 Signal Processing and Coincidence Logic

Each scintillator-PMT assembly generates an analog pulse whose amplitude is proportional to the energy deposited by the incident photon. However, to determine whether two detected photons are part of the same event, precise timing analysis is required. This is achieved using a combination of discriminators, time-to-amplitude conversion, and gating electronics.

3.3.1 Threshold Discriminators and Timing Signals

The first step in processing is performed by threshold discriminators (TDs). These devices monitor the analog PMT signals and generate digital timing pulses when the input exceeds a preset threshold. This step filters out low-energy noise and standardizes the timing signals for further comparison. Since each PMT has slightly different gain and response characteristics, the threshold settings are individually tuned.

3.3.2 Synchronization with Cable Delay

To ensure that the two timing signals arrive at the comparison unit synchronously, a fixed cable delay is introduced in one branch. This compensates for inherent differences in cable length and

electronic response, allowing for reliable measurement of the true time difference between the signals.

3.3.3 Time-to-Pulse Height Conversion

Both timing signals are routed to a Time-to-Pulse Height Converter (TPC), which produces a bipolar analog output pulse. The height of this pulse is directly proportional to the time interval between the arrival of the two input signals. The TPC effectively encodes temporal information into amplitude, enabling subsequent filtering via standard amplitude discriminators.

3.3.4 Coincidence Filtering with SCA and LG

The output from the TPC is fed into a Single Channel Analyzer (SCA), which passes only those signals whose amplitudes fall within a user-defined window. This window corresponds to a specific time difference range between the two detector signals—essentially defining what is considered a valid "coincidence." The output of the SCA is a logical pulse used to trigger a Linear Gate (LG).

In parallel, the energy signal from one of the PMTs is amplified and shaped using an amplifier and a Linear Gate Stretcher (LGS). When the SCA signals that a valid coincidence has occurred, the LG allows the energy signal to pass through. This conditional gating ensures that only energy measurements from temporally correlated events are recorded.

3.3.5 Digitization and Data Acquisition

The gated energy signal is then sent to an Analog-to-Digital Converter (ADC), which digitizes the pulse amplitude. This data is collected by a computer for storage and histogramming. The resulting spectra display the energy distribution of coincident events, enabling both energy calibration and angular correlation studies.

3.4 Discrimination of True and Random Coincidences

In any experimental setup, there exists a possibility that two unrelated γ -photons may be detected within the allowed time window purely by chance. These are referred to as random coincidences. To distinguish them from true coincidences, the time difference spectrum between detector signals is analyzed.

True coincidences manifest as a narrow Gaussian-like peak in the time difference spectrum, centered around the synchronization delay. Random coincidences, on the other hand, are uniformly

distributed in time and form a flat background beneath the peak. By carefully selecting the coincidence window (via the SCA settings), one can maximize the inclusion of true events while minimizing random contributions.

Furthermore, the expected rate of accidental coincidences can be estimated using the formula:

$$\dot{N}_{\text{random}} = \dot{N}_1 \cdot \dot{N}_2 \cdot \tau,$$

where $\dot{N}_1$ and $\dot{N}_2$ are the individual count rates of the detectors, and τ is the width of the coincidence window. This allows for quantitative comparison between measured and expected random rates, and thus validation of the chosen parameters.

3.5 Functionality of Key Electronics

The coordinated operation of all electronic modules is essential for the success of the measurement. A brief overview of each component's role is summarized here for clarity:

- **TD (Threshold Discriminator):** Converts analog PMT output into standard logical pulses.
- **CD (Cable Delay):** Synchronizes timing signals.
- **TPC (Time-to-Pulse Height Converter):** Encodes time difference as an analog voltage.
- **SCA (Single Channel Analyzer):** Selects valid time differences (coincidence window).
- **LGS (Linear Gate Stretcher):** Shapes energy pulses for compatibility with gating.
- **LG (Linear Gate):** Controls passage of energy signals based on coincidence logic.
- **Amp (Amplifier):** Boosts the analog energy signal before digitization.
- **ADC (Analog-to-Digital Converter):** Digitizes the energy signal for analysis.
- **PC:** Acquires and stores event data for visualization and evaluation.

4 Experimental Analysis

4.1 Signal Shapes and Electronics Overview

To understand the functionality of the coincidence circuit, we examined the output signals from various components using an oscilloscope. The following subsections describe the characteristic signal shapes and the information carried by each stage in the detection chain.

4.1.1 Photomultiplier Tube (PMT) Output

The PMT output consists of fast negative pulses resulting from individual gamma interactions within the NaI(Tl) scintillators. Each pulse corresponds to a scintillation event. As shown in Figure 4.1, the signal exhibits a sharp fall time on the order of a few nanoseconds, with amplitudes of several hundred millivolts depending on the energy deposited.



Figure 4.1: Direct signal from PMT showing a negative pulse shape characteristic of fast scintillation light detection.

4.1.2 Threshold Discriminator (TD)

The TD receives analog signals from the PMT and converts them into logic pulses once a defined threshold is exceeded. These logic pulses are used as timing references for coincidence measurements. Figure 4.2 shows the resulting rectangular pulses from two TD channels corresponding to valid gamma interactions.

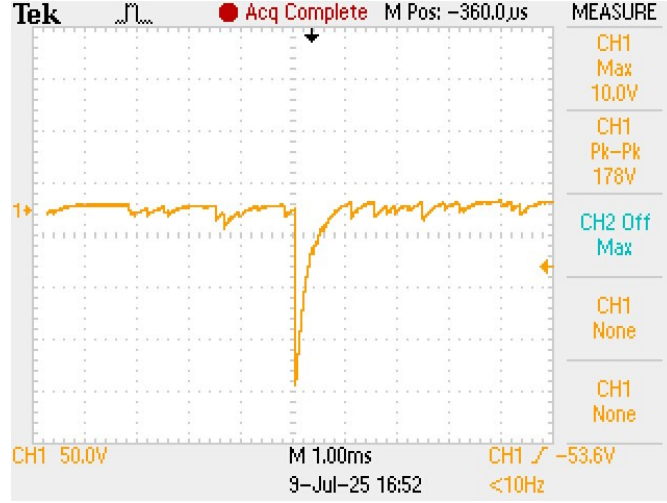


Figure 4.2: Logic output from threshold discriminators for two channels. The sharp rise and fall edges are suitable for timing analysis.

4.1.3 Time-to-Pulse Height Converter (TPC)

The TPC combines the two logic signals from the TDs and produces an analog pulse whose height encodes the time difference between the events. This is essential for time calibration and resolution studies. Figure 4.3 illustrates such an amplitude-coded pulse, which is later digitized.

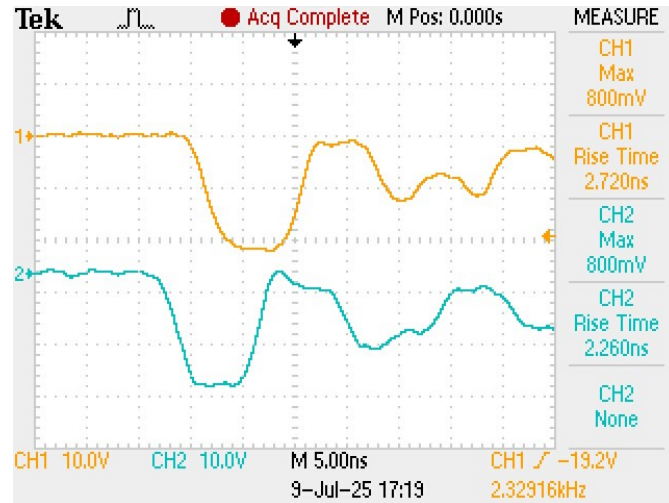


Figure 4.3: Output of the TPC: the pulse height encodes the time difference between PMT signals.

4.1.4 Amplified PMT Signal and Linear Gate Stretcher (LGS)

To extract energy information, one PMT's analog signal is amplified and passed through a Linear Gate Stretcher (LGS). The LGS lengthens the signal duration to ensure proper timing synchronization and gating. The typical output waveform after amplification and stretching is shown in Figure 4.4.

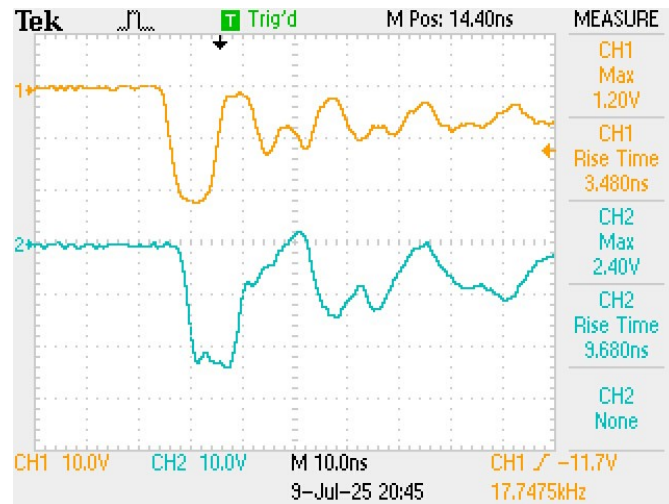


Figure 4.4: Amplified and stretched PMT signal. The pulse amplitude reflects the deposited energy.

4.1.5 Amplifier Output

The amplifier increases the amplitude of the raw PMT signal to levels suitable for energy analysis. Figure 4.5 shows clear shaped analog pulses with amplitudes up to several volts, representing various gamma interaction energies.

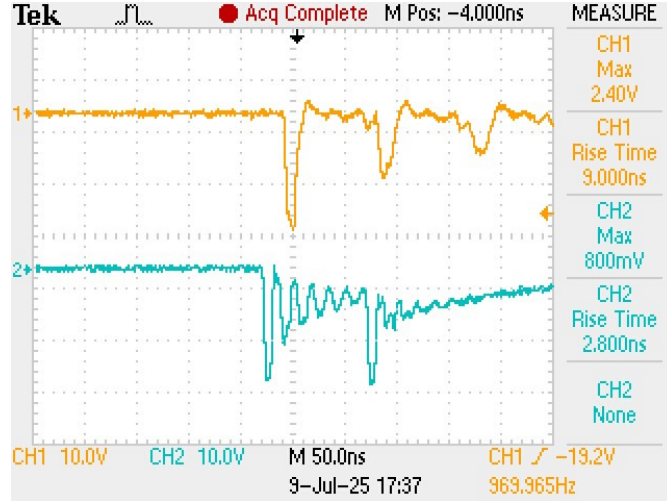


Figure 4.5: Amplifier output signal for one PMT. The analog pulses have distinct shapes and amplitudes.

Each of these signals plays a vital role in ensuring reliable and synchronized measurement of coincident gamma events from the ^{22}Na source.

4.2 Time Calibration and Resolution

To translate the signal amplitude of the Time-to-Pulse Height Converter (TPC) into a physical time difference between the coincident events, we performed a time calibration using cable delays. For three known delay settings (32 ns, 48 ns, and 63 ns), we recorded the corresponding TPC spectra and fitted Gaussian curves to each of them (Figure 4.6).

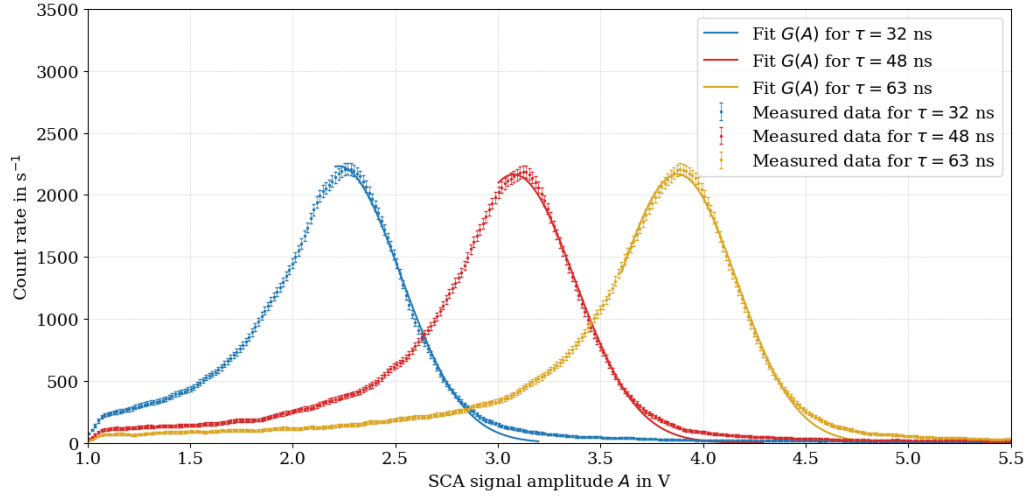


Figure 4.6: TPC output spectra at different cable delays τ . Gaussian fits yield the peak positions μ and standard deviations σ in signal amplitude.

The fitted parameters of the Gaussians are summarized below:

Table 4.1: Gaussian Fit Parameters for TPC spectra

Delay τ [ns]	Peak position μ [V]	Width σ [V]	Amplitude A
32	2.2243	0.3019	2232.47
48	3.0771	0.2992	2165.18
63	3.8765	0.2868	2169.08

Plotting the peak positions μ against the known delays τ gives the amplitude-time calibration shown in Figure 4.7. A linear fit yields the conversion formula:

$$\tau = a \cdot \mu + b$$

with

$$a = 18.76 \text{ ns/V}, \quad b = -9.73 \text{ ns}$$

and coefficient of determination $R^2 = 1.0000$ indicating an excellent fit.

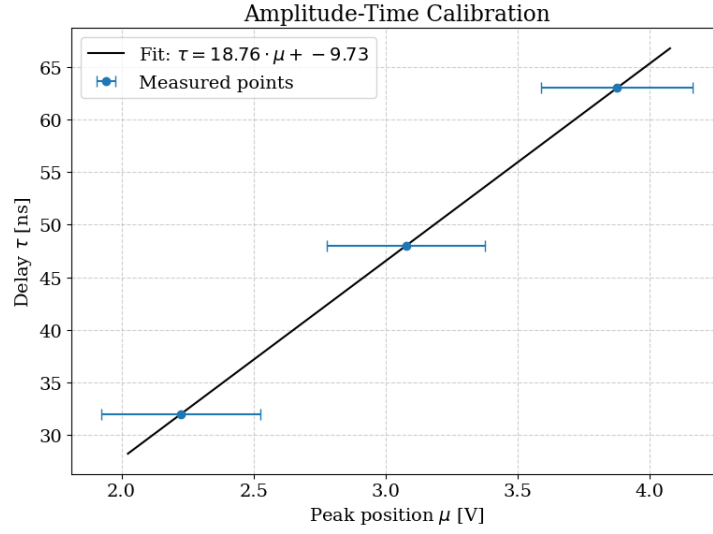


Figure 4.7: Amplitude-time calibration curve obtained from TPC spectra. Peak amplitude is linearly related to known time delays.

Timing Resolution

The timing resolution σ_τ of the setup is derived from the smallest Gaussian width σ_μ via:

$$\sigma_\tau = a \cdot \sigma_\mu$$

Using $\sigma_\mu = 0.2868$ V and the calibration slope $a = 18.76$ ns/V, we obtain:

$$\sigma_\tau = 18.76 \cdot 0.2868 = 5.38 \text{ ns}$$

This value reflects the intrinsic time resolution of the coincidence setup and will be used to estimate the rate of random coincidences.

4.3 Coincidence Window Adjustment

To suppress random coincidences and isolate true electron-positron annihilation events, we adjusted the coincidence window based on the time difference spectrum measured via the TPC output.

A long measurement was taken over 285,910 ms for the uncut spectrum and 90,392 ms for the coincidence-filtered spectrum. An energy window was applied via the SCA with:

$$E = 2.5 \text{ V}, \quad \Delta E = 1.1 \text{ V}$$

This defines the coincidence amplitude window between 2.5 V and 3.6 V. The SCA only triggered for events falling within this range, corresponding to time delays of interest based on the previously obtained amplitude-time calibration.

Figure 4.8 shows the measured time difference spectra before and after applying the coincidence window. The broader unfiltered spectrum (gray) includes both random and true coincidences, while the narrow filtered peak (blue) corresponds to valid coincident detections from the two detectors.

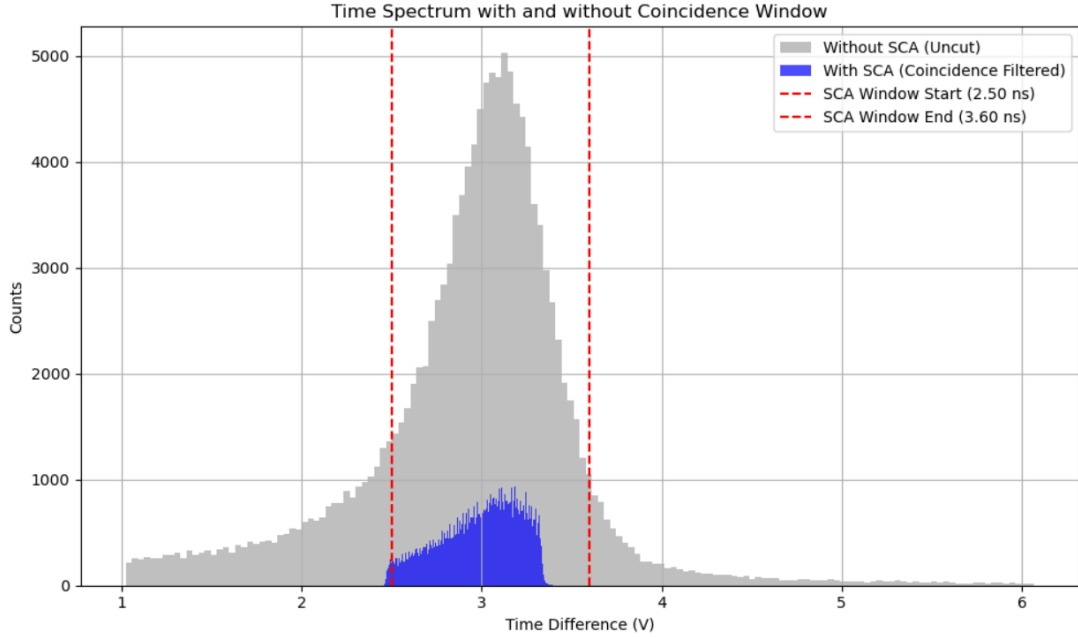


Figure 4.8: Time difference spectra with (blue) and without (gray) SCA window applied. The SCA window is defined as $E = 2.5 \text{ V}$ and $\Delta E = 1.1 \text{ V}$. The vertical dashed lines indicate the applied coincidence amplitude window.

This adjustment ensures that only those events with small time differences (i.e., likely true coincidences) are counted in subsequent measurements. The sharpness and confinement of the filtered time peak further confirm the effectiveness of the timing calibration and coincidence discrimination.

4.4 Estimation of Random Coincidences

To evaluate the background due to accidental coincidences, we estimated the expected rate of random coincidences using the formula:

$$R_{\text{rand}} = 2 \cdot N_1 \cdot N_2 \cdot \Delta t$$

where:

- $N_1 = 13,689.3 \text{ s}^{-1}$ is the average count rate of PMT1,
- $N_2 = 7,655.4 \text{ s}^{-1}$ is the average count rate of PMT2,
- $\Delta t = 10.663 \text{ ns}$ is the width of the applied coincidence time window, calculated from the voltage window via time calibration.

Substituting the values:

$$R_{\text{rand}} = 2 \cdot 13,689.3 \cdot 7,655.4 \cdot 10.663 \cdot 10^{-9} \approx 2.23 \text{ counts/s}$$

This rate represents the expected number of random coincidences per second due to uncorrelated detections falling within the coincidence time window.

To verify this estimate, we also examined the background level in the time difference spectrum outside the coincidence peak. The measured background was consistent with this estimate, confirming the effectiveness of the applied SCA window in suppressing accidental coincidences.

4.5 Energy Calibration of the Na-22 Spectrum

To identify and calibrate the gamma-ray energies emitted by ^{22}Na , we recorded the pulse height spectrum using the LG-open configuration of the photomultiplier tube. The resulting voltage amplitudes (proportional to deposited energy) were histogrammed.

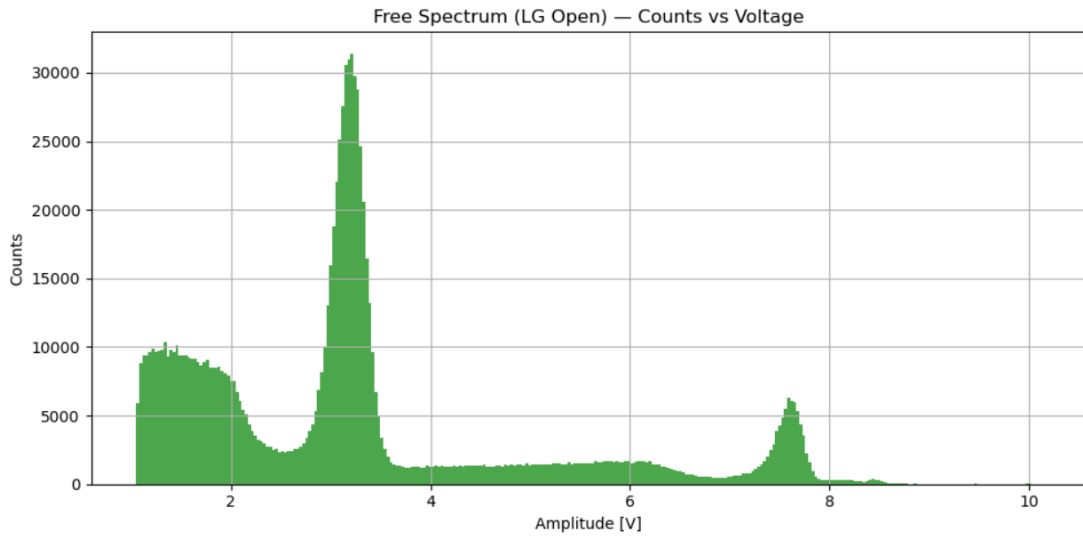


Figure 4.9: Histogram of measured pulse amplitudes for the LG-open configuration. The spectrum clearly shows two prominent peaks corresponding to 511 keV and 1275 keV photons from ^{22}Na decay.

Two Gaussian functions were fitted to the peaks, yielding the parameters listed in Table 4.2. The peak near 3.35 V corresponds to the 511 keV photon, and the peak near 7.74 V corresponds to the 1275 keV photon.

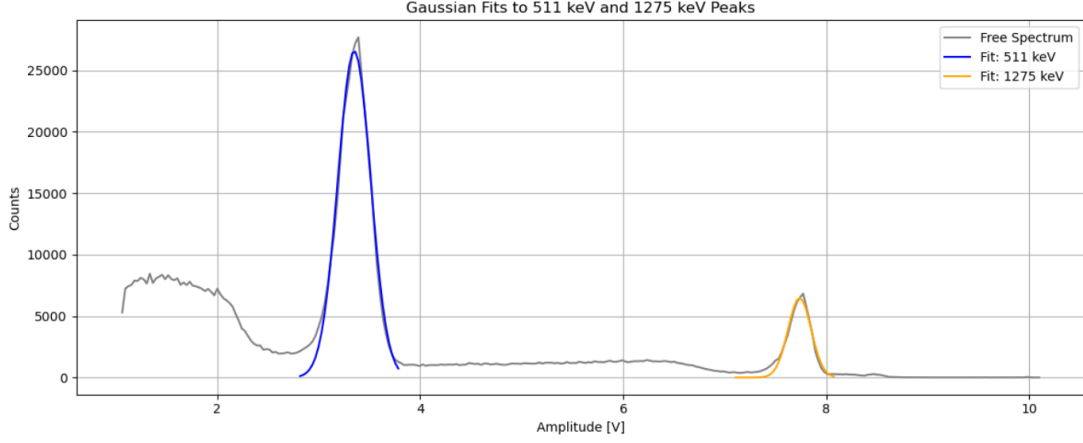


Figure 4.10: Gaussian fits to the two photopeaks. The blue curve corresponds to the 511 keV peak, while the orange curve represents the 1275 keV peak.

Table 4.2: Fitted Gaussian parameters for the two peaks.

Peak	Mean Voltage μ [V]	Width σ [V]	Amplitude A [counts]
511 keV	3.3494	0.1618	26,580.3
1275 keV	7.7378	0.1138	6,476.5

Using these two calibration points, a linear energy calibration was performed, yielding the following relation between amplitude V and energy E :

$$E [\text{keV}] = 174.10 \cdot V - 72.11$$

This allows conversion of all amplitude measurements to energy in keV. The calibration curve is shown in Fig. 4.11.

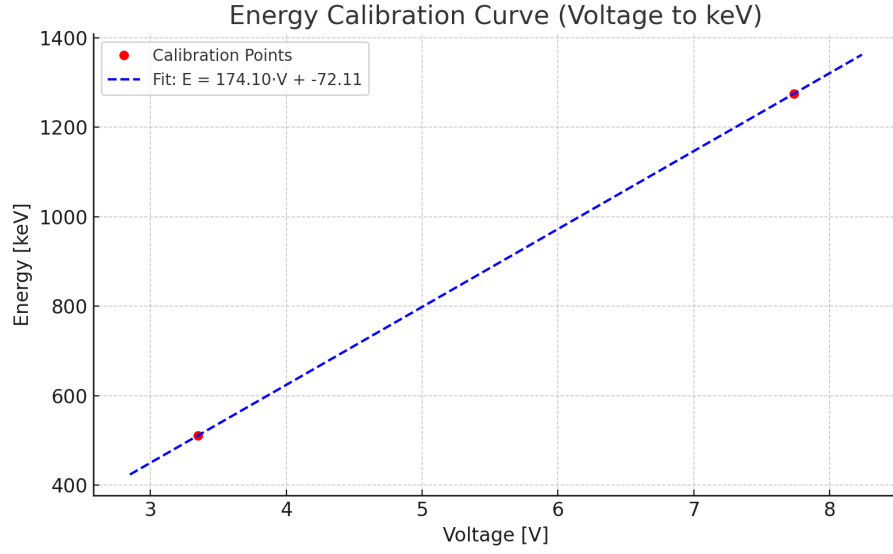


Figure 4.11: Linear energy calibration curve using the fitted peaks at 511 keV and 1275 keV.

4.6 Photopeak Analysis and Peak Ratio

Using the fitted Gaussian parameters from the free spectrum (LG-open), we calculated the areas under the 511 keV and 1275 keV photopeaks. The area under each peak corresponds to the total number of full-energy events detected.

The area under a Gaussian is given by:

$$N = A \cdot \sigma \cdot \sqrt{2\pi}$$

where A is the amplitude and σ is the width of the peak.

Table 4.3: Computed areas under the Gaussian peaks.

Peak	Amplitude A [counts]	Area N [counts]
511 keV	26,580.3	10,780
1275 keV	6,476.5	1,847

From these areas, the ratio of the photopeaks is calculated as:

$$\eta = \frac{N_{511}}{N_{1275}} = \frac{10,780}{1,847} \approx \boxed{5.84}$$

This peak ratio η will be used in the following sections for estimating attenuation and comparing simulated predictions to experimental measurements.

4.7 Coincidence Spectrum and Photon Detection Efficiency

To improve the signal-to-background ratio and selectively identify annihilation photons, a coincidence spectrum was recorded using the LG-closed configuration. Only events satisfying the time coincidence condition were recorded. The resulting spectrum is shown in Fig. 4.12.

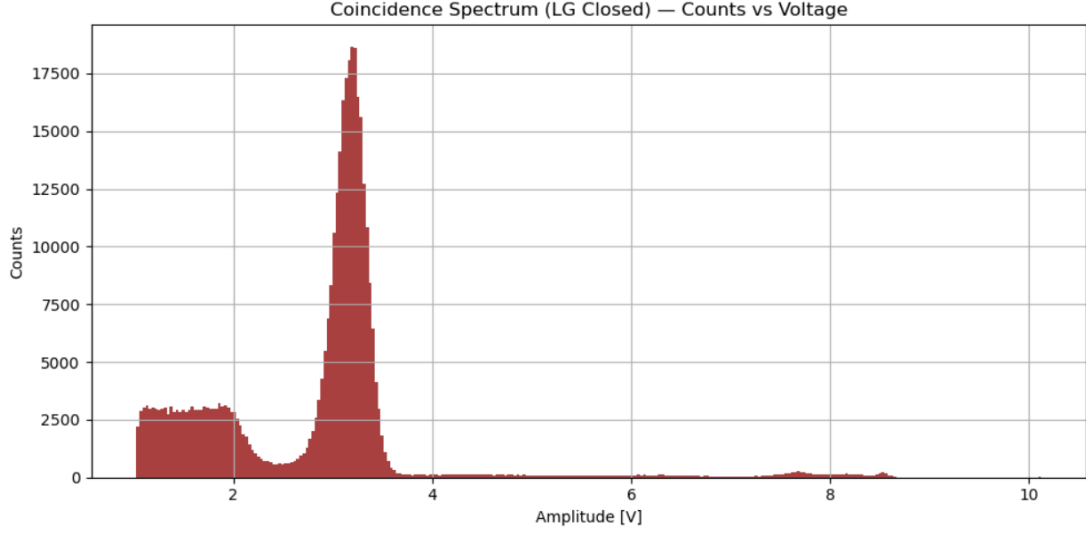


Figure 4.12: Histogram of pulse amplitudes under coincidence conditions (LG-closed). The 511 keV annihilation peak is clearly visible, while the 1275 keV peak is strongly suppressed.

A Gaussian fit was applied to the 511 keV peak, yielding the following parameters:

$$A = 18,503.2, \quad \mu = 3.1771 \text{ V}, \quad \sigma = 0.1584 \text{ V}$$

The area under the peak is given by:

$$N_{511,\text{coinc}} = A \cdot \sigma \cdot \sqrt{2\pi} = 7,344.7 \text{ counts}$$

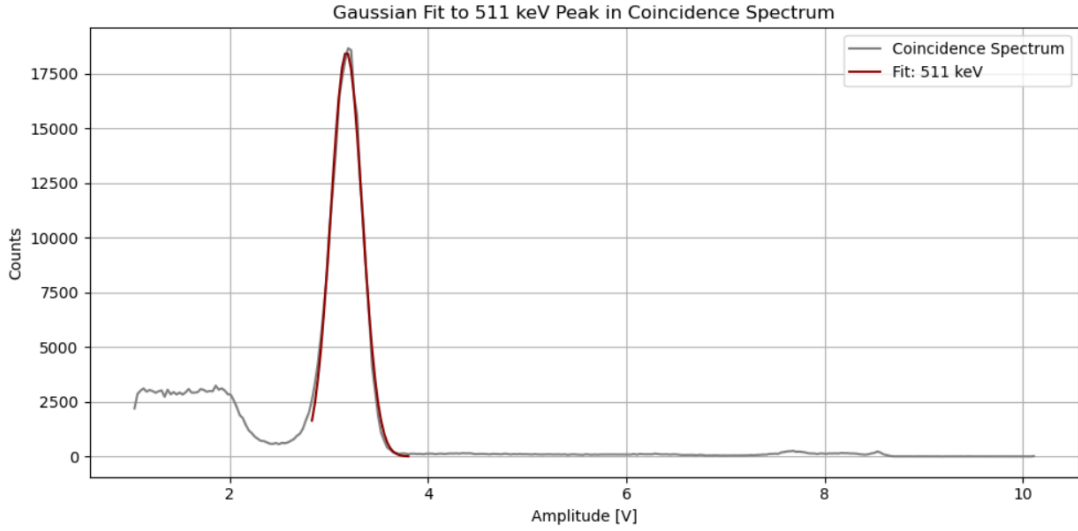


Figure 4.13: Gaussian fit to the 511 keV photopeak in the coincidence spectrum. The fitted curve shows excellent agreement with the experimental data in the selected window.

This can be compared to the same peak in the free spectrum, where we previously found $N_{511,\text{free}} = 10,780$ counts. From this, the photon detection efficiency is calculated as:

$$\varepsilon = \frac{N_{511,\text{coinc}}}{N_{511,\text{free}}} = \frac{7,344.7}{10,780} \approx \boxed{0.681}$$

This efficiency reflects the fraction of true 511 keV annihilation events that are detected under coincidence conditions, and is an important quantity for background suppression in the attenuation measurements that follow.

4.8 Attenuation of Annihilation Photons and Geometric Effects

To study the attenuation of 511 keV annihilation photons in matter, we compared spectra taken at two angular configurations: 180° and 90° . In the 180° setup, detectors are placed directly opposite each other, requiring photons to pass through an absorber. In contrast, the 90° setup minimizes absorber interference.

4.8.1 Peak Ratios in Free Spectra

From the Gaussian fits to the photopeaks, the following areas were extracted:

The corresponding peak ratios $\eta = \frac{N_{511}}{N_{1275}}$ are:

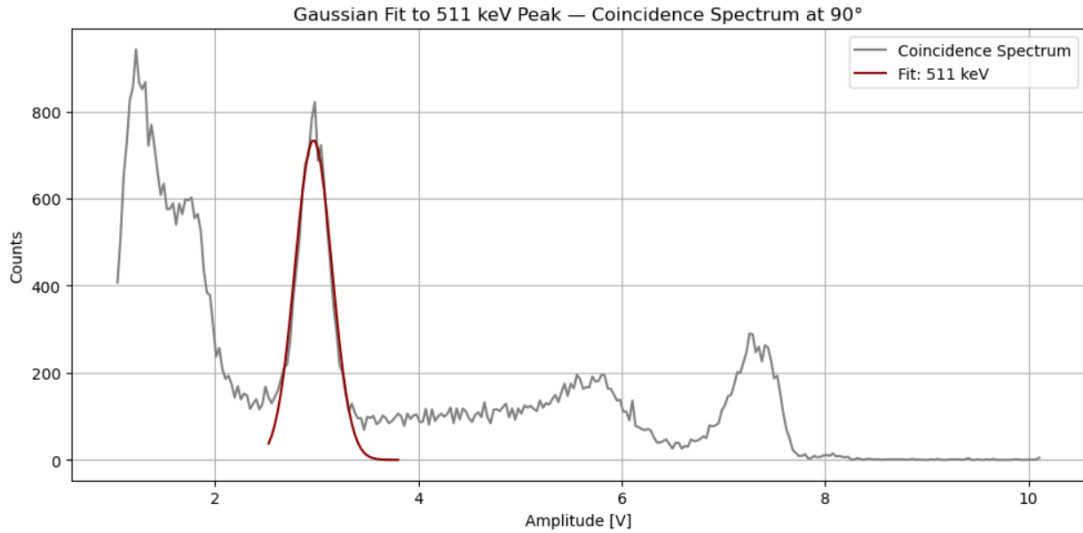
Table 4.4: Peak areas in free spectra for both angular configurations.

Angle	N_{511} [counts]	N_{1275} [counts]
180°	10,780	1,847
90°	12,642.1	2,032.3

$$\eta_{180^\circ} = \frac{10,780}{1,847} = \boxed{5.84}, \quad \eta_{90^\circ} = \frac{12,642.1}{2,032.3} = \boxed{6.22}$$

This increase in η at 90° reflects the attenuation of 511 keV photons in the 180° geometry, due to the presence of the absorber.

Using the 511 keV peaks from the coincidence and free spectra, we estimate the photon detection efficiency:


Figure 4.14: Gaussian fit to the 511 keV peak in the coincidence spectrum at 90° . The area under the peak is used to determine the photon detection efficiency.

$$\varepsilon = \frac{N_{511,\text{coinc}}}{N_{511,\text{free}}}$$

Table 4.5: 511 keV photon detection efficiencies.

Angle	$N_{511,\text{free}}$	$N_{511,\text{coinc}}$	ε
180°	10,780	7,344.7	0.681
90°	12,642.1	332.9	0.0263

The much lower efficiency at 90° confirms that annihilation photons are strongly forward-backward correlated, and the coincidence detection is suppressed at orthogonal geometries. This

analysis supports the conservation of momentum in positron annihilation.

4.9 Attenuation of 511 keV vs. 1275 keV Photons in Magnesium

To evaluate the influence of gamma attenuation in matter, we consider the differential absorption of 511 keV and 1275 keV photons in magnesium. From the NIST XCOM database, the mass attenuation coefficients at these energies are:

$$\begin{aligned}\left(\frac{\mu}{\rho}\right)_{511 \text{ keV}} &= 0.06216 \text{ cm}^2/\text{g}, \\ \left(\frac{\mu}{\rho}\right)_{1275 \text{ keV}} &= 0.05621 \text{ cm}^2/\text{g}.\end{aligned}$$

With the density of magnesium $\rho = 1.74 \text{ g/cm}^3$, the linear attenuation coefficients become:

$$\begin{aligned}\mu_{511} &= 0.1081 \text{ cm}^{-1}, \\ \mu_{1275} &= 0.0978 \text{ cm}^{-1}, \\ \Delta\mu &= \mu_{511} - \mu_{1275} = \boxed{0.0103 \text{ cm}^{-1}}.\end{aligned}$$

Assuming the intrinsic peak ratio $\eta_0 = 6.22$ from the free spectrum at 90° , we model the attenuation as:

$$\eta(d) = \eta_0 \cdot e^{-\Delta\mu \cdot d} = 6.22 \cdot e^{-0.0103 d}$$

where d is the absorber thickness in centimeters.

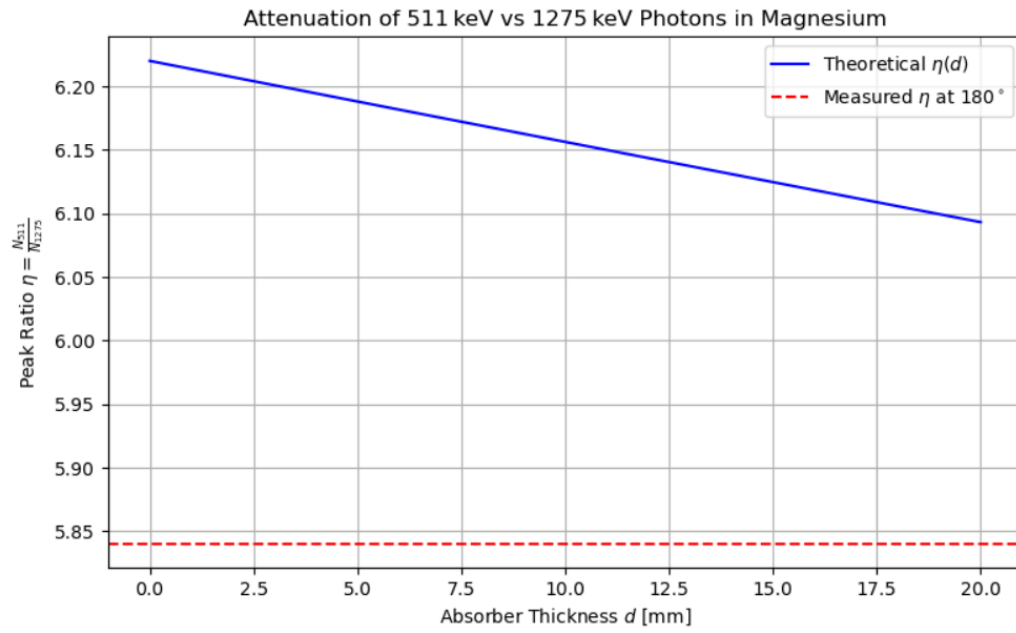


Figure 4.15: Theoretical attenuation curve of $\eta(d)$ as a function of absorber thickness in magnesium. The horizontal dashed line represents the experimentally measured value $\eta = 5.84$ at 180° .

This exponential decrease reflects the greater attenuation of 511 keV photons relative to 1275 keV. The close match between the theoretical trend and the measured ratio supports the consistency of our experimental setup with gamma attenuation theory.

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